Optimal-error signed pseudodistributions for read-once branching programs

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Editorial note. Editable transcription only of formally published CCBY3.0Unported mainPRPD result andfullbearingproofs. Nonexclusive prepublicationnativev4 private aid; publicselected filesexclude allunselectedranges/comments. EntireSections2–6 andAppendixB retained; AppendixA SaksZhou sketch excluded. Originalauthorparameters, norms, signedweights, indices anderrorrecurrences retained without mathematicalrepair oraddedhypotheses. PRPD expectation uses signedrho andis not an ordinary probability distribution; noBPL=L claim. Originalglobal theorem numbering restored andcustomTHM reference type registered. Ordinarybold1 indicator replaces unavailablebbmfont only. Noinput/include/assets. Allproduction/validationYupeng; rootownscompilation.

Allowed standard prerequisites (editorial).

- ROBP transition matrices, averaging samplers/extractors and infinity/max matrix norms. Complete formalSections2–6 allsource-local matrix/robustapproximation/recursive proofs; citedstandardextractor facts andcompleteAppendixB sampler construction retained

Definition 1 (Read-once branching program)**.** *A (n, w)-read-once branching program (ROBP) B is a directed graph on the vertex set $V = \bigcup_{i=0}^{n} V_i$, where each set V_i contains w nodes. Every edge in this directed graph is labeled either* 0 *or* 1*. For every $i < n$, and every node $v \in V_i$, there exists exactly two edges starting from v, one with label* 0 *and the other with label* 1*. Every edge starting from a node in V_i connects to a node in V_{i+1}. We say n is the* length *of B, w is the* width *of B and V_i is the i-th* layer *of B.*

Moreover, there exists exactly one starting state $s \in V_0$, and exactly one accepting state $t \in V_n$. For every $x = (x_1, \ldots, x_n) \in \{0,1\}^n$, we define $B(x) = 1$ if starting from s we will reach t following the edges labeled by $x_1, \ldots, x_n$. Otherwise we define $B(x) = 0$.

It is well-known the computation of a probabilistic Turing machine that uses space S and tosses n coins, on a given input y, can be carried out by a $(n, 2^{O(S)})$-ROBP B_y. In particular, if the string $x \in \{0,1\}^n$ corresponds to the n coin tosses, then $B_y(x)$ is the output of the Turing machine.

A standard derandomization technique is via *pseudorandom generators*. We define this notion for the class of ROBPs.

Definition 2 (Pseudorandom generator)**.** *A function $G : \{0,1\}^s \to \{0,1\}^n$ is a (n, w, ε)-pseudorandom generator (PRG) if for every (n, w)-ROBP B,*

$$\left| \mathop{\mathbb{E}}_{x \in \{0,1\}^n} [B(x)] - \mathop{\mathbb{E}}_{r \in \{0,1\}^s} [B(G(r))] \right| \leq \varepsilon.$$

The seed length *of G is s. G is explicit if G is computable in $O(s)$ space.*

To derandomimze space-bounded computation given an explicit (n, w, ε)-PRG, one can enumerate $B(G(r))$ for every $r \in \{0,1\}^s$ with $O(s)$ additional space to compute an ε-approximation of the quantity $\mathbb{E}_x [B(x)]$.

Nisan [15] constructed a (n, w, ε)-PRG with seed length $O(\log n \cdot \log(nw/\varepsilon))$, which implies $\mathbf{BPL} \subseteq \mathbf{L}^2$. While there is a lot of progress in constructing PRG with better seed length for restricted family of ROBP (see, e.g., [16, 1, 5, 3, 4, 12, 7, 19, 14] and references therein), Nisan's generator and its variants [15, 11, 17] remain the best-known generators in the general case.

0.1 Pseudorandom pseudodistribution

Recently, a beautiful work of Braverman, Cohen and Garg [2] introduced the notion of a *pseudorandom pseudodistribution* (PRPD) that relaxes the definition of a PRG.

Definition 3 (Pseudorandom pseudodistribution)**.** *A pair of functions $(G, \rho) : \{0,1\}^s \to \{0,1\}^n \times \mathbb{R}$ generates a (n, w, ε)-pseudorandom pseudodistribution (PRPD) if for every (n, w)-ROBP B,*

$$\left| \mathop{\mathbb{E}}_{x \in \{0,1\}^n} [B(x)] - \mathop{\mathbb{E}}_{r \in \{0,1\}^s} [\rho(r) \cdot B(G(r))] \right| \leq \varepsilon.$$

We say s is the seed length *of (G, ρ). We say (G, ρ) is k-bounded if $|\rho(x)| \leq k$ for every $x \in \{0,1\}^s$. We say (G, ρ) is explicit if they are computable in space $O(s)$.*

Theorem 4. *There exists an explicit (n, w, ε)-PRPD generator (G, ρ) with seed length*

$$O\left(\log n \cdot \log(nw) \cdot \log\log(nw) + \log(1/\varepsilon)\right),$$

which is $\mathrm{poly}(1/\varepsilon)$*-bounded.*

1 ROBPs and Matrices

We introduce the matrix representation of ROBPs and some related definitions that are useful in the rest of the paper. First, we setup some notation.

Notation: Given two strings x, y, we use $x \| y$ to denote the concatenation of x and y. For every $n \in \mathbb{N}$, we use $[n]$ to denote the set $\{1, 2, \ldots, n\}$. We denote a collection of objects A_i^j with subscript $i \in S$ and superscript $j \in T$ by $[A]_S^T$ for short.

Given a (n, w)-ROBP B with layers $V_0, \ldots, V_n$, we can represent the transition from layer V_{t-1} to V_t by two stochastic matrices M_t^0 and M_t^1 as follows: suppose layer V_j consists of the nodes $\{v_{j,1}, \ldots, v_{j,w}\}$. The entry $(M_t^0)_{i,j} = 1$ if and only if there exist a 0-labeled edge from $v_{t-1,i}$ to $v_{t,j}$ (else $(M_t^0)_{i,j} = 0$).

The matrix M_t^1 is defined similarly according to the edges that labeled 1 between layers V_{t-1} and V_t. More generally, we can also represents multi-step transition by a stochastic matrix. That is, for every $0 \le a \le b \le n$, and every $r = (r_{a+1}, \ldots, r_b) \in \{0,1\}^{b-a}$, we can define

$$M_{a..b}^r = \prod_{t=a+1}^{b} M_t^{r_t}$$

which corresponds to the transition matrix from layer a to layer b following the path labeled by r. Note that every row of $M_{a,b}^r$ contains exactly one 1, and the other entries are 0.

An n-step random walk starting from the first layer can be represented with the following matrix:

$$M_{0..n} = \frac{1}{2^n} \sum_{r \in \{0,1\}^n} M_{0..n}^r = \prod_{t=1}^{n} \frac{1}{2} \left(M_t^0 + M_t^1 \right).$$

By definition of M_t^0, M_t^1 one can observe that the (i,j) entry of $M_{0..n}$ is the probability that a random walk from $v_{0,i} \in V_0$ reaches $v_{n,j} \in V_n$. Therefore, suppose $v_{0,i} \in V_0$ is the starting state of B, $v_{n,j} \in V_n$ is the accepting state of B, then $\mathbb{E}_x [B(x)]$ equals the (i,j) entry of $M_{0..n}$.

Recall that a generator of a (n, w, ε)-PRPD is a pair of function (G, ρ) such that for every (n,w)-ROBP B,

$$\left| \mathop{\mathbb{E}}_{r} [\rho(r) \cdot B(G(r))] - \mathop{\mathbb{E}}_{x \in \{0,1\}^n} [B(x)] \right| \le \varepsilon.$$

Equivalently, for every transition matrices $M_1^0, M_1^1, \ldots, M_n^0, M_n^1$, we have

$$\left\| \mathop{\mathbb{E}}_{r} \left[\rho(r) \cdot M_{0..n}^{G(r)} \right] - M_{0..n} \right\|_{\max} \le \varepsilon,$$

where $\|A\|_{\max}$ denotes $\max_{i,j} |A(i,j)|$.

Therefore it is natural to represents a PRPD (G, ρ) with a mapping $\mathcal{G} : \{0,1\}^s \to \mathbb{R}^{w \times w}$ where $\mathcal{G}(r) = \rho(r) \cdot M_{0..n}^{G(r)}$. More generally, we will use a notation similar to the "matrix bundle sequence" (MBS) introduced in [2] to represent a PRPD.

Definition 5. *Consider a (n,w)-ROBP $[M]_{[n]}^{\{0,1\}}$ and a pair of functions $(G, \rho) : \{0,1\}^{s_{\text{out}}} \times [S_{\text{in}}] \to \{0,1\}^n \times \mathbb{R}$. The* matrix form *of (G, ρ) on $[M]_{[n]}^{\{0,1\}}$ is a mapping $\mathcal{A} : \{0,1\}^{s_{\text{out}}} \times [S_{\text{in}}] \to \mathbb{R}^{w \times w}$ such that for every $x \in \{0,1\}^{s_{\text{out}}}$ and $y \in [S_{\text{in}}]$,*

$$\mathcal{A}(x,y) = \rho(x,y) \cdot M_{0..n}^{G(x,y)}.$$

For every $x \in \{0,1\}^{s_{\text{out}}}$ we abuse the notation and define

$$\mathcal{A}(x) = \mathop{\mathbb{E}}_{y} [\mathcal{A}(x,y)].$$

Besides, we define $\langle \mathcal{A} \rangle = \mathbb{E}_{x,y} [\mathcal{A}(x,y)]$. We say s_{out} is the outer seed length *of $\mathcal{A}$, denoted by $s_{\text{out}}(\mathcal{A})$, and S_{in} is the* inner size *of $\mathcal{A}$, denoted by $S_{\text{in}}(\mathcal{A})$. We also define $s_{\text{in}}(\mathcal{A}) = \lceil \log S_{\text{in}} \rceil$ to be the* inner seed length *of $\mathcal{A}$, and $s(\mathcal{A}) = s_{\text{out}}(\mathcal{A}) + s_{\text{in}}(\mathcal{A})$ to be the* seed length *of $\mathcal{A}$.*

Remark 6. *For every fixed x, the collection $\{\mathcal{A}(x,y) : y \in [S_{\text{in}}]\}$ corresponds to the "matrix bundle" in [2]. This should be treated as a collection of matrices which "realizes" the matrix $\mathcal{A}(x)$. The whole structure $\mathcal{A}$ corresponds to the "matrix bundle sequence" in [2], and should be treated as a uniform distribution over the set $\{\mathcal{A}(x) : x \in \{0,1\}^{s_{\text{out}}}\}$.*

When the ROBP $[M]_{[n]}^{\{0,1\}}$ is clear in the context, we will use the matrix form $\mathcal{A}$ to represent the pseudodistribution (G,ρ) directly. We will apply arithmetic operations on matrices $\mathcal{A}(x)$, and these operations can be easily translated back to operations on pseudodistributions as follows.

Definition 7. *Consider a (n,w)-ROBP $[M]_{[n]}^{\{0,1\}}$, and a pair of function $(F,\sigma) : [S] \to \{0,1\}^n \times \mathbb{R}$. The matrix that is* realized *by (F,σ) on $M_{0..n}$ is $\mathbb{E}_{i\in[S]}\left[\sigma(i)\cdot M_{0..n}^{F(i)}\right]$. We say S is the* size *of (F,σ).*

Scaling the matrix corresponds to scaling the coefficients in the pseudodistribution.

Claim 8. *Consider a (n,w)-ROBP $[M]_{[n]}^{\{0,1\}}$, let A be a matrix realized by matrix bundle (F_A,σ_A) on $M_{0..n}$. Then cA is realized by a matrix bundle (F'_A,σ'_A) of size S_A s.t. $F'_A = F_A$ and $\sigma'_A(x) = c\sigma_A(x)$ for every $x \in [S]$.*

The summation on matrices corresponds to re-weighting and union on pseudodistributions.

Claim 9. *Consider a (n,w)-ROBP $[M]_{[n]}^{\{0,1\}}$, let A be a matrix realized by matrix bundle (F_A,σ_A) of size S_A on $M_{0..n}$ and B be a matrix realized by matrix bundle (F_B,σ_B) of size S_B on $M_{0..n}$. Then $A+B$ is realized by a matrix bundle (F',σ') of size S_A+S_B on $M_{0..n}$ s.t.*

$$F'(x) = \begin{cases} F_A(x) & \text{if } x \le S_A \\ F_B(x-S_A) & \text{if } x > S_A \end{cases} \text{ and } \sigma'(x) = \begin{cases} \frac{S_A+S_B}{S_B}\cdot\sigma_A(x) & \text{if } x \le S_A \\ \frac{S_A+S_B}{S_B}\cdot\sigma_B(x-S_A) & \text{if } x > S_A \end{cases}$$

The multiplication on matrices corresponds to concatenation of pseudodistributions.

Claim 10. *Consider a (n,w)-ROBP $[M]_{[n]}^{\{0,1\}}$, let A be a matrix realized by matrix bundle (F_A,σ_A) of size S_A on $M_{0..n/2}$ and B be a matrix realized by matrix bundle (F_B,σ_B) of size S_B on $M_{n/2..n}$. Fix a bijection $\pi : [S_A]\times[S_B] \to [S_A\cdot S_B]$. Then AB is realized by a matrix bundle (F',σ') of size $S_A\cdot S_B$ s.t. for every $a \in [S_A], b \in [S_B]$,*

$$F'(\pi(a,b)) = F_A(a)\|F_B(b) \text{ and } \sigma'(\pi(a,b)) = \sigma(a)\cdot\sigma(b).$$

2 Proof Overview

In this section we give an outline of our construction and proof. In Section 2.1, we briefly recap how a sampler is used in [2] to achieve better seed length in the small-error regime. We discuss our construction ideas in Section 2.2.

2.1 The sampler argument

Nisan's generator and its variants recursively use a lemma of the following form.

Lemma 11. *Consider a (n, w)-ROBP $[M]_{[n]}^{\{0,1\}}$. Let $\mathcal{A}$ be the matrix form of a distribution on $M_{0..n/2}$, and $\mathcal{B}$ be the matrix form of a distribution on $M_{n/2..n}$. Suppose $s(\mathcal{A}) = s(\mathcal{B}) = s$. Then there exists a distribution whose matrix form $\mathcal{C}$ on $M_{0..n}$ of seed length $s + O(\log(w/\delta))$ such that*

$$\|\langle \mathcal{C} \rangle - \langle \mathcal{A} \rangle \langle \mathcal{B} \rangle\|_{\max} \leq \delta.$$

This lemma is usually achieved with a pseudorandom object. For example, the INW generator [11] uses a bipartite expander with degree $\mathrm{poly}(w/\delta)$ to construct the distribution $\mathcal{C}$ in the above lemma. That is, for every edge (x, y) in the expander G, they add $\mathcal{A}(x)\mathcal{B}(y)$ into $\mathcal{C}$. A similar lemma can also be obtained with universal hash functions [15] or seeded extractors [17]. By recursively constructing good approximations of $M_{0..n/2}$ and $M_{n/2..n}$ and applying Theorem 11, one can obtain a PRG which has seed length $O(\log n \cdot \log(nw/\varepsilon))$ (δ is taken to be ε/n because of a union bound). Observe that in such constructions, one needs to pay $O(\log(1/\varepsilon))$ (in seed length) per level of recursion.

The crucial idea in [2] is to amortize this cost over all $\log n$ levels. What makes this possible is the following argument, which we will refer to as the *sampler argument*. First we define the notion of an averaging sampler.

Definition 12. *A function $g : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ is an (ε, δ)-(averaging) sampler if for every function $f : \{0,1\}^m \to [0,1]$,*

$$\Pr_{x \in \{0,1\}^n} \left[\left| \mathop{\mathbb{E}}_{s \in \{0,1\}^d} [f(g(x,s))] - \mathop{\mathbb{E}}_{y \in \{0,1\}^m} [f(y)] \right| \leq \varepsilon \right] \geq 1 - \delta.$$

The crucial observation in [2] is that if one uses a sampler to prove Theorem 11, the error actually scales with the norm of one of the matrix forms.

Lemma 13 ([2])**.** *Consider a (n, w)-ROBP with matrix representation $[M]_{[n]}^{\{0,1\}}$. Let $\mathcal{A}$ and $\mathcal{B}$ be (pseudo)distributions in matrix form on $M_{0..n/2}$ and $M_{n/2..n}$ respectively. Let $n = s_{\mathrm{out}}(\mathcal{A})$, $m = s_{\mathrm{out}}(\mathcal{B})$. Suppose $\forall x \in \{0,1\}^n, \|\mathcal{A}(x)\| \leq 1$ and $\forall y \in \{0,1\}^m, \|\mathcal{B}(y)\| \leq 1$. Let $g : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ be a (ε, δ) sampler. Then there exists a (pseudo)distribution $\mathcal{C}$ such that*

$$\|\langle C \rangle - \langle A \rangle \langle B \rangle\| \leq O\left(w^2 \left(\delta + \varepsilon \mathop{\mathbb{E}}_x [\|\mathcal{A}(x)\|]\right)\right).$$

Besides, $\mathcal{C}$ has outer seed length $n = s_{\mathrm{out}}(\mathcal{A})$, and for every $x \in \{0,1\}^n$,

$$\mathcal{C}(x) = \mathop{\mathbb{E}}_s [\mathcal{A}(x)\mathcal{B}(g(x,s))].$$

Note that $s_{\mathrm{in}}(\mathcal{C}) = s_{\mathrm{in}}(\mathcal{A}) + s_{\mathrm{in}}(\mathcal{B}) + d$.

The intuition behind this approximation is as follows. If we want to compute the matrix product precisely, we take every $\mathcal{A}(x)$ and multiply it with $\mathbb{E}_y [\mathcal{B}(y)]$. However, with the help of sampler, we can use x as our seed to select some samples from $\mathcal{B}$, and take their average as an estimate of $\mathbb{E}_y [\mathcal{B}(y)]$. The error of this approximation comes in two different way. For those x which are not good

choices of a seed for the sampler, the samples chosen with such an x can deviate from the average arbitrarily. However, only δ fraction of x can be bad, so they incur at most δ error. The second kind of error is the estimation error between average of samples $\mathbb{E}_s[\mathcal{B}(g(x,s))]$ and the real average $\mathbb{E}_y[\mathcal{B}(y)]$, which can be at most ε. Since this gets multiplied with $\mathcal{A}(x)$, this kind of error actually scales with $\|\mathcal{A}(x)\|$. Although the first kind of error (which is δ) does not benefit from $\|\mathcal{A}\|$ being small, in [2] they observe that, the parameter δ has almost no influence on the seed length in some cases. To discuss this more precisely, we first recall explicit constructions of samplers.

Lemma 14 ([18, 8])**.** *For every $\delta, \varepsilon > 0$ and integer m, there exists a space efficient (ε, δ)-sampler $f : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ s.t. $d = O(\log\log(1/\delta) + \log(1/\varepsilon))$ and $n = m + O(\log(1/\delta)) + O(\log(1/\varepsilon))$.*

Note that in Theorem 13, $s(\mathcal{C}) = s(\mathcal{A}) + d + s_{\text{in}}(\mathcal{B})$. Therefore if $n \geq m + O(\log(1/\delta)) + O(\log(1/\varepsilon))$, δ has almost no impact on the seed length.

To use the above ideas, it boils down to working with matrices with small norm, and making sure that every multiplication is "unbalanced" enough so that δ has no impact. [2] applies a delicate telescoping sum trick (which they called "delta sampler") to divide an ε-approximation into a base approximation with $1/\text{poly}(n)$ error and several "correcting terms" which have small norm. By carefully choosing the samplers and discarding all the non-necessary terms, they roughly ensure the following properties: first, a matrix with large seed length must have small norm; second, every matrix multiplication is unbalanced enough so that δ has no impact on the seed length.

With these properties and the sampler argument, they show that the total seed length is bounded by $\tilde{O}(\log(1/\varepsilon) + \log n \log(nw))$.

2.2 Our construction

In executing the ideas sketched above, the construction and analysis in [2] turns out to be quite complicated and involved. One thing which complicates the construction and analysis is its *bottom-up* nature. That is, when multiplying two terms, they create more terms with the telescoping sum trick. Moreover, in the telescoping sum trick one needs to choose the parameters of each sampler very carefully to make sure the seed length of each term does not exceed its "smallness".

Our first step toward a simpler construction is the following *top-down* formula, which we will apply recursively to compute an approximation of $M_{0..n}$:

Lemma 15. *Let $\|\cdot\|$ be a sub-multiplicative matrix norm, and A, B be two matrices s.t. $\|A\|, \|B\| \leq 1$. Let $k \in \mathbb{N}$ and $\gamma < 1$. For every $0 \leq i \leq k$, let A_i be a γ^{i+1}-approximation of A, and let B_i be a γ^{i+1}-approximation of B. Then*

$$\sum_{i=0}^{k} A_i B_{k-i} - \sum_{i=0}^{k-1} A_i B_{k-1-i}$$

is a $((k+2)\gamma^{k+1} + (k+1)\gamma^{k+2})$-approximation of AB.

Proof. We have,

$$\left\| (\sum_{i=0}^{k} A_i B_{k-i} - \sum_{i=0}^{k-1} A_i B_{k-1-i}) - AB \right\|$$

$$
\begin{aligned}
&= \left\| \sum_{i=0}^{k} (A - A_i)(B - B_{k-i}) - \sum_{i=0}^{k-1} (A - A_i)(B - B_{k-1-i}) + (A_k - A)B + A(B_k - B) \right\| \\
&\leq \sum_{i=0}^{k} \|A - A_i\| \cdot \|B - B_{k-i}\| + \sum_{i=0}^{k-1} \|A - A_i\| \cdot \|B - B_{k-1-i}\| \\
&+ \|A_k - A\| \cdot \|B\| + \|A\| \cdot \|B_k - B\| \\
&\leq (k+2)\gamma^{k+1} + (k+1)\gamma^{k+2}
\end{aligned}
$$

□

This formula shares an important property with the BCG construction: we never need a γ^k-approximation (which implies large seed length) on both sides simultaneously. The benefit of our top-down formula is that we are treating the PRPD as one object instead of the sum of many different terms. One obvious effect of such treatment is we don't need to analyze the "smallness" of each term and the accuracy of the whole PRPD separately.

In this top-down formula, we do not explicitly maintain small-norm matrices as in [2]. However, observe that in the proof of Theorem 15, we are using the fact that $A_k - A$ is a small norm matrix. Our goal is to apply the sampler argument (Theorem 13) on these "implicit" small-norm matrices. The following is our main technical lemma.

Lemma 16 (main lemma, informal)**.** *Let $A, B \in \mathbb{R}^{w \times w}$, $k \in \mathbb{N}$ and $\gamma < 1$. Suppose for every $i \leq k$ there exists pseudodistribution $\mathcal{A}_i, \mathcal{B}_i$ such that $\mathbb{E}_x \left[\|\mathcal{A}_i(x) - A\|\right] \leq \gamma^{i+1}$, $\mathbb{E}_x \left[\|\mathcal{B}_i(x) - B\|\right] \leq \gamma^{i+1}$, and $\|\mathcal{A}_i(x)\|, \|\mathcal{B}_i(x)\| \leq 1$ for every x. Then there exists a pseudo-distribution $\mathcal{C}_k$ such that*

$$
\mathop{\mathbb{E}}_{x} \left[\|\mathcal{C}_k(x) - AB\| \leq O(\gamma)^{k+1}\right],
$$

where $\mathcal{C}_k(x) = \sum_{i+j=k} A_{x,i} B_{x,j} - \sum_{i+j=k-1} A_{x,i} B_{x,j}$. $A_{x,i}$ and $B_{x,i}$ are defined as follows.

- *If $i > \lceil k/2 \rceil$, $A_{x,i} = \mathcal{A}_i(x)$ and $B_{x,i} = \mathcal{B}_i(x)$.*
- *If $i \leq \lceil k/2 \rceil$, $A_{x,i} = \mathbb{E}_s \left[\overline{\mathcal{A}_i}(g_i(x,s))\right]$ and $B_{x,i} = \mathbb{E}_s \left[\overline{\mathcal{B}_i}(g_i(x,s))\right]$, where g_i is a $(\gamma^{i+1}, \gamma^{k+1})$-sampler, and $\overline{\mathcal{A}_i}, \overline{\mathcal{B}_i}$ denote the "flattened" form of $\mathcal{A}_i$ and $\mathcal{B}_i$.*

We leave the explanation of "flattened" for later and explain the intuition behind the lemma first. Our goal is to construct $\mathcal{C}_k$ such that $\mathcal{C}_k(x)$ is a good approximation of AB *on average* over x. We know that $\mathcal{A}_i$ and $\mathcal{B}_i$ are γ^{i+1}-approximation of A and B on average. Our hope is to use x to draw samples A_i and B_i from $\mathcal{A}_i$ and $\mathcal{B}_i$, and apply the formula in Theorem 15 to get a good approximation of AB. In particular, a natural choice would be setting $A_{x,i} = \mathcal{A}_i(x)$ and $B_{x,i} = \mathcal{B}_i(x)$ for every $i \leq k$. However, if there exists a term $A_{x,i}B_{x,j}$ such that $A_{x,i}$ and $B_{x,j}$ are both bad approximation for a large enough fraction of x, we cannot guarantee to get a $O(\gamma^{k+1})$-approximation on average.

To avoid the above case, for every $i \leq \lceil k/2 \rceil$ we use a sampler to approximate $\langle \mathcal{A}_i \rangle$ and $\langle \mathcal{B}_i \rangle$. This ensure that the chosen samples $A_{x,i}$ and $B_{x,i}$ are good *with high probability*. This guarantees that in each term $A_{x,i}B_{x,j}$, at least one of $A_{x,i}$ or $B_{x,j}$ will be a good choice with high probability over x. If $A_{x,i}$ is a

good choice with high probability, we can apply the average-case guarantee on $B_{x,i}$ to get an average-case guarantee for $\mathcal{C}_k$, and vice versa. (Indeed, this is the sampler argument.) Therefore we can ensure that $\mathcal{C}_k(x)$ is good on average. Note that we only apply a sampler on $\mathcal{A}_i$ (or $\mathcal{B}_i$) when $i \le \lceil k/2 \rceil$, which means $\mathcal{A}_i$ (or $\mathcal{B}_i$) has small seed length. Therefore we don't need to add too much redundant seed to make the sampler argument work.

In executing the above sketched idea, we run into the following problem: in each multiplication, the inner seed on both sides aggregates to the upper level. If we start with pseudodistributions with non-zero inner seed in the bottom level, the inner seed would become $\Omega(n)$ in the topmost level. Therefore we need a way to limit the aggregation of inner seed.

In [2], they run into a similar problem. To deal with this, they apply a different multiplication rule, "outer product", in some special cases to deal with this. However, the outer product does not seem applicable in our construction. Nevertheless, we observe that whenever we use a sampler to select matrix $A_{x,i}$, we only care about whether $\langle \mathcal{A}_i \rangle$ is close to A, and we don't need most of $\mathcal{A}_i(x)$ to be close to A anymore. Therefore we will "flatten" $\mathcal{A}_i$ whenever we apply a sampler. That is, recall that each $\mathcal{A}_i(x)$ is realized by the average of some matrices, $\mathbb{E}_y [\mathcal{A}_i(x, y)]$. We define the flattened form of $\mathcal{A}_i$, denoted by $\overline{\mathcal{A}_i}$, such that $\overline{\mathcal{A}_i}(x \| y) = \mathcal{A}_i(x, y)$. Observe that $\langle \overline{\mathcal{A}_i} \rangle = \langle \mathcal{A}_i \rangle$ and $s_{\text{in}}(\overline{\mathcal{A}_i}) = 0$. This guarantees that the inner seed length of $\mathcal{A}_i$ will not aggregate in $\mathcal{C}_k$. Moreover, while the flattening will increase the outer seed length of $\overline{\mathcal{A}_i}$, this is almost for free since we only flatten $\mathcal{A}_i$ when $i \le \lceil k/2 \rceil$, i.e. when $\mathcal{A}_i$ has relatively small seed length. As a result, this operation also helps us save a $O(\log \log(1/\varepsilon))$ factor in the seed length.

We conclude by briefly discussing the seed length analysis. First note that we set $\gamma = 1/\text{poly}(n)$ to make sure that the error is affordable after a union bound. Now consider the inner seed length. Consider a term $A_i B_j$ such that $i \ge j$. In this term, part of the inner seed of $\mathcal{C}$ is passed to $\mathcal{A}_i$, and the other is used for the sampler on B_j. Since the seed length of the sampler only needs to be as large as the "precision gap" between $\mathcal{A}_i$ and $\mathcal{C}_k$, the inner seed length of $\mathcal{C}_k$ can be maintained at roughly $O(k \log(1/\gamma)) = O(\log(1/\varepsilon))$. However, after each multiplication, there's actually a $O(\log(nw/\gamma)) = O(\log(nw))$ additive overhead. Note that this is necessary since the $k = 0$ case degenerates to the INW generator. Therefore after $\log n$ levels of recursion, the inner seed length will be $O(\log(1/\varepsilon) + \log n \cdot \log(nw))$.

Besides, we also need the outer seed length of $\mathcal{C}_k$ to be long enough so that we can apply a sampler on $\mathcal{A}_{\lceil k/2 \rceil}$ and $\mathcal{B}_{\lceil k/2 \rceil}$. The seed length caused by approximation accuracy ε can be bounded similarly as the inner seed length. However, the $O(\log n \cdot \log(nw))$ inner seed length will be added to the outer seed length several times, because of the flattening operation. Nevertheless, since we only do flattening for $\mathcal{A}_i$ and $\mathcal{B}_i$ where $i \le \lceil k/2 \rceil$, this ensures that the flattening operation happens at most $\log k$ times. So the total outer seed length will be bounded by $O(\log(1/\varepsilon) + \log k \cdot \log n \cdot \log(nw)) = O(\log(1/\varepsilon) + \log \log(1/\varepsilon) \cdot \log n \cdot \log(nw))$, which is bounded by $O(\log(1/\varepsilon) + \log \log(nw) \cdot \log n \cdot \log(nw))$ since $O(\log(1/\varepsilon))$ is the dominating term when $\log(1/\varepsilon) \ge \log^3(nw)$.

3 Preliminaries

3.1 Averaging samplers

Definition 17. *A function $g : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ is a (ε, δ)* (averaging) sampler *if for every function $f : \{0,1\}^m \to [0,1]$,*

$$\Pr_{x \in \{0,1\}^n}\left[\left|\mathop{\mathbb{E}}_{s \in \{0,1\}^d}[f(g(x,s))] - \mathop{\mathbb{E}}_{y \in \{0,1\}^m}[f(y)]\right| \le \varepsilon\right] \ge 1 - \delta.$$

It's easy to show that samplers also work for f with general range by scaling and shifting.

Claim 18. *Let $g : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ be a (ε, δ)-sampler, and let $\ell < r \in \mathbb{R}$. Then for every $f : \{0,1\}^m \to [\ell, r]$,*

$$\Pr_{x \in \{0,1\}^n}\left[\left|\mathop{\mathbb{E}}_{s \in \{0,1\}^d}[f(g(x,s))] - \mathop{\mathbb{E}}_{y \in \{0,1\}^m}[f(y)]\right| \le \varepsilon(r-\ell)\right] \ge 1 - \delta.$$

Proof. Let f' be the function such that $f'(y) = (f(y) - \ell)/(r - \ell)$. Observe that the range of f' is in $[0,1]$. By definition of sampler,

$$\Pr_{x \in \{0,1\}^n}\left[\left|\mathop{\mathbb{E}}_{s \in \{0,1\}^d}[f'(g(x,s))] - \mathop{\mathbb{E}}_{y \in \{0,1\}^m}[f'(y)]\right| \le \varepsilon\right] \ge 1 - \delta.$$

By multiplying $(r - \ell)$ on both sides of the inequality inside the probability above we prove the claim. □

In our construction, we will use the following sampler which is explicitly computable with small space.

Lemma 19 ([18, 8])**.** *For every $\delta, \varepsilon > 0$ and integer m, there exists a (ε, δ)-sampler $f : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ s.t. $d = O(\log\log(1/\delta) + \log(1/\varepsilon))$ and $n = m + O(\log(1/\delta)) + O(\log(1/\varepsilon))$. Moreover, for every x, y, $f(x,y)$ can be computed in space $O(m + \log(1/\delta) + \log(1/\varepsilon))$.*

Remark 20. *The original sampler in [18] has a restriction on ε. Such a restriction will cause a $2^{O(\log^*(nw/\varepsilon))}$ factor in our construction, as in [2]. However, [18] pointed out that the restriction is inherited from the extractor in [21], which breaks down when the error is extremely small. As observed in [8], this restriction can be removed by plugging in a more recent extractor construction in [9]. Note that there exists a space-efficient implementation of [9] in [13], so the resulting sampler is also space-efficient. For completeness we include a proof in Section B.*

3.2 Matrix norms

As in [2], we will use the infinity norm in this paper.

Definition 21. *For every matrix $A \in \mathbb{R}^{w \times w}$, $\|A\| = \max_i \sum_j |A_{i,j}|$.*

We record some well known properties of the infinity norm.

Claim 22. *Let $A, B \in \mathbb{R}^{w \times w}$, $c \in \mathbb{R}$. Then*

- $\|cA\| = |c| \|A\|$
- $\|A\| + \|B\| \leq \|A + B\|$
- $\|AB\| \leq \|A\| \|B\|$
- $\max_{i,j} |A_{i,j}| \leq \|A\|$
- *If A is stochastic, then* $\|A\| = 1$

Note that for any (n, w)-ROBP represented by $w \times w$ matrices $M_{[n]}^{\{0,1\}}$, $\|M_{i..j}\| = 1$ for every $0 \leq i \leq j \leq n$.

4 Approximate Matrix Multiplication via Samplers

In this section we formally prove the sampler arguments which will be used in our construction. Our proof strategy resembles that of [2], with the following two crucial differences. First, we will define two different notions of "smallness" for our flattening idea. Second, in our construction we need the case where we use samplers to select matrices on both sides (Theorem 27).

We will consider mappings $\mathcal{A} : \{0,1\}^n \to \mathbb{R}^{w \times w}$ which correspond to the implicit small norm matrices we discussed in the previous section. Borrowing notation from Theorem 5, we use $\langle \mathcal{A} \rangle$ to denote $\mathbb{E}_x [\mathcal{A}(x)]$. First we define two different norms for the mapping $\mathcal{A}$. The *robust norm* is similar to the notion of "smallness" in [2], i.e. the average of norm of $\mathcal{A}(x)$, while the *norm* of $\mathcal{A}$ is simply the norm of $\langle \mathcal{A} \rangle$, i.e. the norm of average of $\mathcal{A}(x)$.

Definition 23. *For every function $\mathcal{A} : \{0,1\}^n \to \mathbb{R}^{w \times w}$, we define the* norm *of $\mathcal{A}$ to be $\|\mathcal{A}\| = \left\|\mathbb{E}_{x \in \{0,1\}^n} [\mathcal{A}(x)]\right\|$, and the* robust norm *of $\mathcal{A}$ to be $\|\mathcal{A}\|_r = \mathbb{E}_{x \in \{0,1\}^n} [\|\mathcal{A}(x)\|]$. Besides, we define the weight of $\mathcal{A}$ to be $\mu(\mathcal{A}) = \max_x \|\mathcal{A}(x)\|$.*

Claim 24. $\|\mathcal{A}\| \leq \|\mathcal{A}\|_r \leq \mu(\mathcal{A})$.

Proof. $\|\mathcal{A}\| \leq \|\mathcal{A}\|_r$ is by sub-additivity of $\|\cdot\|$, and $\|\mathcal{A}\|_r \leq \mu(\mathcal{A})$ since $\|\mathcal{A}\|_r$ is the average of values no larger than $\mu(\mathcal{A})$. □

Next we show a simple lemma which will be used later. That is, a sampler for functions with range $[0, 1]$ is also a sampler for matrix-valued functions, where the error is measured with infinity norm.

Lemma 25. *For every function $\mathcal{A} : \{0,1\}^m \to \mathbb{R}^{w \times w}$ and every (ε, δ)-sampler $g : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$,*

$$\Pr_{x \in \{0,1\}^n} \left[\left\| \mathop{\mathbb{E}}_{s \in \{0,1\}^d} [\mathcal{A}(g(x,s))] - \langle \mathcal{A} \rangle \right\| \leq 2w\mu(\mathcal{A})\varepsilon \right] \geq 1 - w^2\delta.$$

Proof. Let $\mathcal{E}(y) = \mathcal{A}(y) - \langle \mathcal{A} \rangle$. For every $i, j \in [w]$, observe that

$$\max_y \mathcal{E}(y)_{i,j} - \min_y \mathcal{E}(y)_{i,j} = \max_y \mathcal{A}(y)_{i,j} - \min_y \mathcal{A}(y)_{i,j}$$

By the property of sampler it follows that

$$\Pr_{x\in\{0,1\}^n}\left[\left|\mathop{\mathbb{E}}_{s}\left[\mathcal{E}(g(x,s))_{i,j}\right]\right| \le 2\varepsilon\mu(\mathcal{A})\right] \ge 1-\delta.$$

Using a union bound,

$$\Pr_{x\in\{0,1\}^n}\left[\forall i,j\in[w], \left|\mathop{\mathbb{E}}_{s}\left[\mathcal{E}(g(x,s))_{i,j}\right]\right| \le 2\varepsilon\mu(\mathcal{A})\right] \ge 1-w^2\delta.$$

Thus by definition of the infinity norm, we can conclude that

$$\Pr_{x\in\{0,1\}^n}\left[\left\|\mathop{\mathbb{E}}_{s\in\{0,1\}^d}\left[\mathcal{E}(g(x,s))\right]\right\| \le 2w\mu(\mathcal{A})\varepsilon\right] \ge 1-w^2\delta.$$

which by sub-additivity of $\|\cdot\|$ implies

$$\Pr_{x\in\{0,1\}^n}\left[\left\|\mathop{\mathbb{E}}_{s}\left[\mathcal{A}(g(x,s))\right]\right\| \le \|\mathcal{A}\| + 2w\mu(\mathcal{A})\varepsilon\right] \ge 1-w^2\delta.$$

□

Corollary 26. *For every function $\mathcal{A}:\{0,1\}^m\to\mathbb{R}^{w\times w}$ and every (ε,δ)-sampler $g:\{0,1\}^n\times\{0,1\}^d\to\{0,1\}^m$,*

$$\Pr_{x\in\{0,1\}^n}\left[\left\|\mathop{\mathbb{E}}_{s\in\{0,1\}^d}\left[\mathcal{A}(g(x,s))\right]\right\| \le \|\mathcal{A}\| + 2w\mu(\mathcal{A})\varepsilon\right] \ge 1-w^2\delta.$$

Proof. By sub-additivity of $\|\cdot\|$, $\left\|\mathbb{E}_{s\in\{0,1\}^d}\left[\mathcal{A}(g(x,s))\right]-\langle A\rangle\right\| \le 2w\mu(\mathcal{A})\varepsilon$ implies $\left\|\mathbb{E}_{s\in\{0,1\}^d}\left[\mathcal{A}(g(x,s))\right]\right\| \le \|\langle\mathcal{A}\rangle\| + 2w\mu(\mathcal{A})\varepsilon$. The claim now directly follows from Theorem 25. □

Now we introduce three different matrix multiplication rules. The first one is applying a sampler on both sides, and the second and third are applying sampler on only one side.

Lemma 27 (symmetric product)**.** *Consider $\mathcal{A}:\{0,1\}^n\to\mathbb{R}^{w\times w}$ and $\mathcal{B}:\{0,1\}^m\to\mathbb{R}^{w\times w}$. Let $f:\{0,1\}^k\times\{0,1\}^{d_A}\to\{0,1\}^n$ be a (δ,ε_A) sampler, and $g:\{0,1\}^k\times\{0,1\}^{d_B}\to\{0,1\}^m$ be a (δ,ε_B) sampler. Then*

$$\mathop{\mathbb{E}}_{z}\left[\left\|\mathop{\mathbb{E}}_{x,y}\left[\mathcal{A}(f(z,x))\mathcal{B}(g(z,y))\right]\right\|\right] \le 2w^2\delta\mu(\mathcal{A})\mu(\mathcal{B}) + \left(\|\mathcal{A}\| + 2w\mu(\mathcal{A})\varepsilon_A\right)\left(\|\mathcal{B}\| + 2w\mu(\mathcal{B})\varepsilon_B\right).$$

Proof. Let

$$E_A = \left\{z : \left\|\mathop{\mathbb{E}}_{x}\left[\mathcal{A}(f(z,x))\right]\right\| > \|\mathcal{A}\| + 2w\mu(\mathcal{A})\varepsilon_A\right\},$$

and

$$E_B = \left\{z : \left\|\mathop{\mathbb{E}}_{y}\left[\mathcal{B}(g(z,y))\right]\right\| > \|\mathcal{B}\| + 2w\mu(\mathcal{B})\varepsilon_B\right\}.$$

Define $E = E_A\cup E_B$. By Lemma 26 and union bound, $\Pr_z[z\in E] < 2w^2\delta$. Therefore

$$\mathop{\mathbb{E}}_{z}\left[\left\|\mathop{\mathbb{E}}_{x,y}\left[\mathcal{A}(f(z,x))\mathcal{B}(g(z,y))\right]\right\|\right] = \Pr[z\in E]\mathop{\mathbb{E}}_{z\in E}\left[\left\|\mathop{\mathbb{E}}_{x,y}\left[\mathcal{A}(f(z,x))\mathcal{B}(g(z,y))\right]\right\|\right]$$

$$
\begin{aligned}
&+ \Pr[z \notin E] \mathop{\mathbb{E}}_{z \notin E} \left[\left\| \mathop{\mathbb{E}}_{x} [\mathcal{A}(f(z,x))] \mathop{\mathbb{E}}_{y} [\mathcal{B}(g(z,y))] \right\| \right] \\
&\leq 2w^2 \delta \mu(\mathcal{A}) \mu(\mathcal{B}) + \mathop{\mathbb{E}}_{z \notin E} \left[\left\| \mathop{\mathbb{E}}_{x} [\mathcal{A}(f(z,x))] \right\| \left\| \mathop{\mathbb{E}}_{y} [\mathcal{B}(g(z,y))] \right\| \right] \\
&\leq 2w^2 \delta \mu(\mathcal{A}) \mu(\mathcal{B}) + (\|\mathcal{A}\| + 2w\mu(\mathcal{A})\varepsilon_A) (\|\mathcal{B}\| + 2w\mu(\mathcal{B})\varepsilon_B) .
\end{aligned}
$$

The second last inequality is by the fact that $\|\cdot\|$ is non-negative and sub-multiplicative. □

Lemma 28 (left product)**.** *Consider $\mathcal{A} : \{0,1\}^k \to \mathbb{R}^{w \times w}$ and $\mathcal{B} : \{0,1\}^m \to \mathbb{R}^{w \times w}$. Let $g : \{0,1\}^k \times \{0,1\}^{d_B} \to \{0,1\}^m$ be a (δ, ε_B) sampler. Then*

$$
\mathop{\mathbb{E}}_{z} \left[\left\| \mathop{\mathbb{E}}_{y} [\mathcal{A}(z)\mathcal{B}(g(z,y))] \right\| \right] \leq w^2 \delta \mu(\mathcal{A}) \mu(\mathcal{B}) + \|\mathcal{A}\|_r (\|\mathcal{B}\| + 2w\mu(\mathcal{B})\varepsilon_B) .
$$

Proof. Let

$$
E = \left\{ z : \left\| \mathop{\mathbb{E}}_{y} [\mathcal{B}(g(z,y)] \right\| > \|\mathcal{B}\| + 2w\mu(\mathcal{B})\varepsilon_B \right\} .
$$

By Lemma 26, $\Pr_z[z \in E] < w^2\delta$. Therefore

$$
\begin{aligned}
\mathop{\mathbb{E}}_{z} \left[\left\| \mathop{\mathbb{E}}_{y} [\mathcal{A}(z)\mathcal{B}(g(z,y))] \right\| \right] &= \Pr[z \in E] \mathop{\mathbb{E}}_{z \in E} \left[\left\| \mathop{\mathbb{E}}_{y} [\mathcal{A}(z)\mathcal{B}(g(z,y))] \right\| \right] \\
&\quad + \Pr[z \notin E] \mathop{\mathbb{E}}_{z \notin E} \left[\left\| \mathop{\mathbb{E}}_{y} [\mathcal{A}(z)\mathcal{B}(g(z,y))] \right\| \right] \\
&\leq w^2 \delta \mu(\mathcal{A}) \mu(\mathcal{B}) + \Pr[z \notin E] \cdot \mathop{\mathbb{E}}_{z \notin E} \left[\|\mathcal{A}(z)\| \left\| \mathop{\mathbb{E}}_{y} [\mathcal{B}(g(z,y))] \right\| \right] \\
&\leq w^2 \delta \mu(\mathcal{A}) \mu(\mathcal{B}) + \Pr[z \notin E] \mathop{\mathbb{E}}_{z \notin E} [\|\mathcal{A}(z)\|] \cdot (\|\mathcal{B}\| + 2w\mu(\mathcal{B})\varepsilon_B) \\
&\leq w^2 \delta \mu(\mathcal{A}) \mu(\mathcal{B}) + \|\mathcal{A}\|_r (\|\mathcal{B}\| + 2w\mu(\mathcal{B})\varepsilon_B) .
\end{aligned}
$$

The third last inequality is by sub-multiplicativity of $\|\cdot\|$, the second last inequality is by non-negativity of $\|\cdot\|$, and the last inequality is by the fact that

$$
\Pr[z \notin E] \cdot \mathop{\mathbb{E}}_{z \notin E} [\|\mathcal{A}(z)\|] = \mathop{\mathbb{E}}_{z} [\|\mathcal{A}(z)\| \cdot \mathbf{1}(z \notin E)] \leq \|\mathcal{A}\|_r .
$$

□

Lemma 29 (right product)**.** *Consider $\mathcal{A} : \{0,1\}^k \to \mathbb{R}^{w \times w}$ and $\mathcal{B} : \{0,1\}^m \to \mathbb{R}^{w \times w}$. Let $f : \{0,1\}^k \times \{0,1\}^{d_A} \to \{0,1\}^n$ be a (δ, ε_A) sampler. Then*

$$
\mathop{\mathbb{E}}_{z} \left[\left\| \mathop{\mathbb{E}}_{x} [\mathcal{A}(f(z,x))\mathcal{B}(z)] \right\| \right] \leq w^2 \delta \mu(\mathcal{A}) \mu(\mathcal{B}) + (\|\mathcal{A}\| + 2w\mu(\mathcal{A})\varepsilon_A) \|\mathcal{B}\|_r .
$$

Proof. Let

$$
E = \left\{ z : \left\| \mathop{\mathbb{E}}_{x} [\mathcal{A}(f(z,x)] \right\| > \|\mathcal{A}\| + 2w\mu(\mathcal{A})\varepsilon_A \right\} .
$$

By Lemma 26, $\Pr_z[z \in E] < w^2\delta$. Therefore

$$
\mathop{\mathbb{E}}_{z} \left[\left\| \mathop{\mathbb{E}}_{x} [\mathcal{A}(f(z,x))\mathcal{B}(z)] \right\| \right] = \Pr[z \in E] \mathop{\mathbb{E}}_{z \in E} \left[\left\| \mathop{\mathbb{E}}_{x} [\mathcal{A}(f(z,x))\mathcal{B}(z)] \right\| \right]
$$

$$
\begin{aligned}
&+ \Pr[z \notin E] \mathop{\mathbb{E}}_{z \notin E} \left[\left\| \mathop{\mathbb{E}}_{x} [\mathcal{A}(f(z,x))\mathcal{B}(z)] \right\| \right] \\
&\leq w^2 \delta \mu(\mathcal{A}) \mu(\mathcal{B}) + \Pr[z \notin E] \cdot \mathop{\mathbb{E}}_{z \notin E} \left[\left\| \mathop{\mathbb{E}}_{x} [\mathcal{A}(f(z,x))] \right\| \|\mathcal{B}(z)\| \right] \\
&\leq w^2 \delta \mu(\mathcal{A}) \mu(\mathcal{B}) + (\|\mathcal{A}\| + 2w\mu(\mathcal{A})\varepsilon_A) \cdot \Pr[z \notin E] \mathop{\mathbb{E}}_{z \notin E} [\|\mathcal{B}(z)\|] \\
&\leq w^2 \delta \mu(\mathcal{A}) \mu(\mathcal{B}) + (\|\mathcal{A}\| + 2w\mu(\mathcal{A})\varepsilon_A) \|\mathcal{B}\|_r .
\end{aligned}
$$

□

5 Main Construction

In this section we show our main construction and prove its correctness. We first introduce several definitions.

Definition 30. *For every mapping $\mathcal{A} : \{0,1\}^n \to \mathbb{R}^{w \times w}$ and every matrix $A \in \mathbb{R}^{w \times w}$, we define $\mathcal{A} - A$ to be the mapping s.t. $(\mathcal{A} - A)(x) = \mathcal{A}(x) - A$.*

Definition 31. *Consider $A \in \mathbb{R}^{w \times w}$ and $\mathcal{A} : \{0,1\}^n \to \mathbb{R}^{w \times w}$. $\mathcal{A}$ is a ε-*approximator *of A if $\|\mathbb{E}_x[\mathcal{A}(x)] - A\| \leq \varepsilon$, i.e. $\|\mathcal{A} - A\| \leq \varepsilon$. $\mathcal{A}$ is a ε-*robust approximator *of A if $\mathbb{E}_x[\|\mathcal{A}(x) - A\|] \leq \varepsilon$, i.e. $\|\mathcal{A} - A\|_r \leq \varepsilon$.*

Now we define a robust PRPD. Note that a (n, w, ε)-robust PRPD (G, ρ) is also a $\mu(G, \rho)$-bounded (n, w, ε)-PRPD.

Definition 32. *$(G, \rho) : \{0,1\}^{s_{\text{out}}} \times \{0,1\}^{s_{\text{in}}} \times [\mu] \to \{0,1\}^n \times \mathbb{R}$ is a (n, w, ε)-*robust PRPD *if for every (n, w)-ROBP and its matrix representation $[M]_{[n]}^{\{0,1\}}$ the following holds. Let $\mathcal{A} : \{0,1\}^{s_{\text{out}}} \times \{0,1\}^{s_{\text{in}}} \to \mathbb{R}^{w \times w}$ denote the mapping*

$$
\mathcal{A}(x, y) = \mathop{\mathbb{E}}_{i \in [\mu]} \left[\rho(x, y, i) \cdot M_{0..n}^{G(x,y,i)} \right].
$$

- *Every $\rho(x, y, i)$ is either μ or $-\mu$. In other word, $\mathcal{A}(x, y)$ is the summation of transition matrices with coefficient ± 1.*
- *Let $\widehat{\mathcal{A}}$ denote the mapping $\widehat{\mathcal{A}}(x) = \mathbb{E}_y[\mathcal{A}(x, y)]$. Then $\widehat{\mathcal{A}}$ is a ε-robust approximator for $M_{0..n}$.*

We say μ is the weight of (G, ρ), denoted by $\mu(G, \rho)$. s_{out} is the outer seed length *of (G, ρ), denoted by $s_{\text{out}}(G, \rho)$. s_{in} is the* inner seed length *of (G, ρ), denoted by $s_{\text{in}}(G, \rho)$. We write $s(G, \rho) = s_{\text{out}}(G, \rho) + s_{\text{in}}(G, \rho)$ for short. We say (G, ρ) is explicit if it can be computed in $O(s(G, \rho))$ space.*

We say $\mathcal{A}$ is the matrix form *of (G, ρ) on $M_{0..n}$, and the definition of $s_{\text{out}}, s_{\text{in}}, \mu$ on (G, ρ) also apply to $\mathcal{A}$. We say $\widehat{\mathcal{A}}$ is the* robust matrix form *of (G, ρ) on $M_{0..n}$.*

Remark 33. *The above definition is similar to Theorem 5, but each matrix $\mathcal{A}(x, y)$ is realized with μ matrices instead of one matrix. These μ matrices will never be separated even after flattening. We do this in order to ensure that the matrix form always take bit-strings as input. This ensures that we can increase the outer and inner seed length of $\mathcal{A}$ arbitrarily: we can construct the new mapping $\mathcal{A}' : \{0,1\}^{s'_{\text{out}}} \times \{0,1\}^{s'_{\text{in}}}$ such that $\mathcal{A}'(x, y) = \mathcal{A}(x_p, y_p)$ where x_p*

is the length-$s_{\text{out}}(\mathcal{A})$ prefix of x and y_p is the length-$s_{\text{in}}(\mathcal{A})$ prefix of y. In other word, $\mathcal{A}'$ computes the output only with prefix of necessary length of the input, and ignore the remaining bits. It is easy to verify that $\mathcal{A}'$ is also the matrix form of a (n, w, ε)-robust PRPD.

The following is some additional basic properties about robust PRPD and its flattened form.

Claim 34. *Let $(G, \rho) : \{0,1\}^{s_{\text{out}}} \times \{0,1\}^{s_{\text{in}}} \times [\mu] \to \{0,1\}^n \times \mathbb{R}$ be a (n, w, ε)-*robust PRPD*. *For every (n, w)-ROBP $M_1^0, M_1^1, \ldots, M_n^0, M_n^1$ the following holds.*

- *Let $\widehat{\mathcal{A}}$ be the robust matrix form of (G, ρ) on $M_{0..n}$. Then $\mu(\widehat{\mathcal{A}}) \le \mu(G, \rho)$.*
- *Let $\mathcal{A}$ denote the matrix form of (G, ρ) on $M_{0..n}$. Let $\overline{\mathcal{A}} : \{0,1\}^{s_{\text{out}}+s_{\text{in}}} \to \mathbb{R}^{w \times w}$ denote the mapping $\overline{\mathcal{A}}(x\|y) = \mathcal{A}(x, y)$. We say $\overline{\mathcal{A}}$ is the* flattened matrix form *of (G, ρ) on $M_{0..n}$. Then $\overline{\mathcal{A}}$ is an ε-approximator for $M_{0..n}$, and $\mu(\overline{\mathcal{A}}) \le \mu(G, \rho)$.*

Proof. Recall that for every string $r \in \{0,1\}^n$, $\|M_{0..n}^r\| = 1$. By sub-additivity of $\|\cdot\|$ we have $\|\mathcal{A}(x, y)\| \le \mu(G, \rho)$ for every x, y, which implies $\mu(\overline{\mathcal{A}}) \le \mu(G, \rho)$. By sub-additivity and scalibility of $\|\cdot\|$, we have $\mu(\mathcal{A}') \le \mu(\mathcal{A})$. To show that $\overline{\mathcal{A}}$ is a ε-approimxator of $M_{0..n}$, observe that $\mathcal{A}'$ is also an ε-approximator of $M_{0..n}$ by Theorem 24, and note that $\langle \mathcal{A} \rangle = \langle \mathcal{A}' \rangle$. □

Now we prove our main lemma. The following lemma allows us to construct robust PRPDs for $(2m, w)$ ROBPs from robust PRPDs for (m, w) ROBPs, without increasing the seed length too much. We will recursively apply this lemma for $\log n$ levels to get a (n, w, ε)-robust PRPD. The basic idea is as described in Theorem 16.

Lemma 35. *Suppose there exists $s_{\text{out}}, s_{\text{in}}$ such that the following conditions hold.*

- *For every $0 \le i \le k$, there exists a (m, w, γ^{i+1})-robust PRPD (G_i, ρ_i) s.t. $\mu(G_i, \rho_i) \le \binom{m-1}{i}$ and $s_{\text{out}}(G, \rho) \le s_{\text{out}}$. Moreover, for every $0 \le i \le \lceil k/2 \rceil$, $s(G_i, \rho_i) \le s_{\text{out}}$.*
- *For every $i \le \lceil k/2 \rceil$, there exists a (ε_i, δ)-sampler $g_i : \{0,1\}^{s_{\text{out}}} \times \{0,1\}^{d_i} \to \{0,1\}^{s(G_i, \rho_i)}$, where $\varepsilon_i \le \gamma^{i+1}/(w \cdot \binom{m-1}{i})$ and $\delta \le \gamma^{k+1}/(w^2 \cdot \binom{2m-1}{i})$.*
- *For every $i \ge j \ge 0$ s.t. $i + j \le k$, if $j \le i \le \lceil k/2 \rceil$, then $d_i + d_j \le s_{\text{in}}$. If $i > \lceil k/2 \rceil$, then $s_{\text{in}}(G_i, \rho_i) + d_j \le s_{\text{in}}$.*

Then there exists a $(2m, w, (11\gamma)^{k+1})$-robust PRPD (G, ρ) s.t. $s_{\text{out}}(G, \rho) = s_{\text{out}}$, $s_{\text{in}}(G, \rho) = s_{\text{in}}$ and $\mu(G, \rho) \le \binom{2m-1}{k}$.

Proof. Fix any $(2m, w)$-ROBP with matrix representation $M_{[2m]}^{\{0,1\}}$. Let $A = M_{0..m}$ and $B = M_{m..2m}$. For every $0 \le i \le k$, let $\mathcal{A}_i, \widehat{\mathcal{A}}_i, \overline{\mathcal{A}}_i$ denote the matrix form, robust matrix form and flattened matrix form of (G, ρ) on $M_{0..m}$ respectively. Let $\mathcal{B}_i, \widehat{\mathcal{B}}_i, \overline{\mathcal{B}}_i$ denote the matrix form, robust matrix form and flattened matrix form of (G, ρ) on $M_{m..2m}$ respectively. By definition, $\widehat{\mathcal{A}}_i$ and $\widehat{\mathcal{B}}_i$ are γ^{i+1}-robust approximator for A and B respectively. By Theorem 34, $\overline{\mathcal{A}}_i$ and $\overline{\mathcal{B}}_i$ are γ^{i+1}-approximator for A and B respectively. Moreover, we will increase

the outer seed length of $\mathcal{A}_i$ and $\mathcal{B}_i$ to match the length of the given input when necessary. (See Theorem 33)

Now for every x, y we define a mapping $\mathcal{C}_k : \{0,1\}^{s_{\text{out}}} \times \{0,1\}^{s_{\text{in}}} \to \mathbb{R}^{w \times w}$ as follows. Note that $\mathcal{C}_k$ corresponds to the matrix form of (G, ρ) on $M_{0..2m}$.

(1) For every $0 \le i \le \lceil k/2 \rceil$, let a_i be the prefix of y of length d_i and b_i be the suffix of y of length d_i. Define $A_{x,y,i} = \overline{\mathcal{A}_i}(g_i(x, a_i))$ and $B_{x,y,i} = \overline{\mathcal{B}_i}(g_i(x, b_i))$.

(2) For every $\lceil k/2 \rceil < i \le k$, let a_i be the prefix of y of length $s_{\text{in}}(\mathcal{A}_i)$ and b_i be the suffix of y of length $s_{\text{in}}(\mathcal{B}_i)$. Define $A_{x,y,i} = \mathcal{A}_i(x, a_i)$ and $B_{x,y,i} = \mathcal{B}_i(x, b_i)$.

(3) Define $\mathcal{C}_k(x, y) = \sum_{i+j=k} A_{x,y,i} B_{x,y,j} - \sum_{i+j=k-1} A_{x,y,i} B_{x,y,j}$.

Note that for every $i + j \le k$, prefix a_i and suffix b_j of y never overlap.

By expanding every $A_{x,y,i} B_{x,y,j}$ term with distributive law, we can see that each small term in $A_{x,y,i} B_{x,y,j}$ has coefficient ± 1, which satisfies the first condition of robust PRPD. Moreover, the total number of terms after expanding is

$$\mu(\mathcal{C}_k) \le \sum_{i+j=k} \binom{m-1}{i} \cdot \binom{m-1}{j} + \sum_{i+j=k-1} \binom{m-1}{i} \cdot \binom{m-1}{j} = \binom{2m-1}{k}.$$

It remains to show that $\mathcal{C}_k$ satisfies the second condition of robust PRPD, i.e. $\mathbb{E}_y [\mathcal{C}_k(x, y)]$ is a good approximation of $M_{0..2m} = AB$ on average over x. Observe that

$$\begin{aligned}
\mathop{\mathbb{E}}_x \left[\left\| \mathop{\mathbb{E}}_y [\mathcal{C}_k(x,y)] - AB \right\| \right] &= \mathop{\mathbb{E}}_x \left[\left\| \mathop{\mathbb{E}}_y [\mathcal{C}_k(x,y) - AB] \right\| \right] \\
&\le \sum_{i+j=k} \mathop{\mathbb{E}}_x \left[\left\| \mathop{\mathbb{E}}_y [(A_{x,y,i} - A)(B_{x,y,j} - B)] \right\| \right] \\
&+ \sum_{i+j=k-1} \mathop{\mathbb{E}}_x \left[\left\| \mathop{\mathbb{E}}_y [(A_{x,y,i} - A)(B_{x,y,j} - B)] \right\| \right] \\
&+ \mathop{\mathbb{E}}_x \left[\left\| \mathop{\mathbb{E}}_y [(A_{x,y,k} - A)B] \right\| \right] + \mathop{\mathbb{E}}_x \left[\left\| \mathop{\mathbb{E}}_y [A(B_{x,y,k} - B)] \right\| \right],
\end{aligned}$$

by decomposing $\mathcal{C}_k(x, y) - AB$ with the equation in the proof of Theorem 15 and applying sub-additivity of $\|\cdot\|$.

First we consider the last two terms. Since $\|B\| = 1$, by sub-multiplicativity we have

$$\mathop{\mathbb{E}}_x \left[\left\| \mathop{\mathbb{E}}_y [(A_{x,y,k} - A)B] \right\| \right] \le \mathop{\mathbb{E}}_x \left[\left\| \mathop{\mathbb{E}}_y [A_{x,y,k} - A] \right\| \right].$$

Now consider two cases. If $k \ge 2$, then

$$\mathop{\mathbb{E}}_x \left[\left\| \mathop{\mathbb{E}}_y [A_{x,y,k} - A] \right\| \right] = \mathop{\mathbb{E}}_x \left[\left\| \widehat{\mathcal{A}}_k(x) - A \right\| \right] \le \gamma^{k+1}$$

by definition. If $k < 2$, then

$$\mathop{\mathbb{E}}_x \left[\left\| \mathop{\mathbb{E}}_y [A_{x,y,k} - A] \cdot B \right\| \right] = \mathop{\mathbb{E}}_x \left[\left\| \mathop{\mathbb{E}}_{a_k} [\overline{\mathcal{A}_k}(g_k(x, a_k)) - A] \right\| \cdot B \right].$$

Apply Theorem 29 on $\overline{\mathcal{A}_k} - A$ and the dummy mapping $\mathcal{B}$ s.t. $\mathcal{B}(x) = B$ for every x, we can derive that the above formula is bounded by $w^2\delta\binom{m-1}{k} + 3\gamma^{i+1}$. For the term $\mathbb{E}_x\left[\|\mathbb{E}_y\left[A(B_{x,y,k} - B)\right]\|\right]$ we can get the same bound with a similar proof.

Now consider the terms in the form $\mathbb{E}_x\left[\|\mathbb{E}_y\left[(A_{x,y,i} - A)(B_{x,y,j} - B)\right]\|\right]$. First consider the case $i, j \le \lceil k/2 \rceil$. Then

$$\begin{aligned}
&\mathop{\mathbb{E}}_x\left[\left\|\mathop{\mathbb{E}}_y\left[(A_{x,y,i} - A)(B_{x,y,j} - B)\right]\right\|\right] \\
&= \mathop{\mathbb{E}}_x\left[\left\|\mathop{\mathbb{E}}_{a_i}\left[\overline{\mathcal{A}_i}(g_i(x,a_i)) - A\right]\mathop{\mathbb{E}}_{b_j}\left[\overline{\mathcal{B}_k}(g_j(x,b_j)) - B\right]\right\|\right] \text{ (since } a_i, b_j \text{ don't overlap)} \\
&\le 2w^2\delta \cdot \binom{m-1}{i} \cdot \binom{m-1}{j} + 9\gamma^{i+j+2}. \text{ (by Theorem 27)}
\end{aligned}$$

Next consider the case $i > \lceil k/2 \rceil, j \le \lceil k/2 \rceil$. Then

$$\begin{aligned}
&\mathop{\mathbb{E}}_x\left[\left\|\mathop{\mathbb{E}}_y\left[(A_{x,y,i} - A)(B_{x,y,j} - B)\right]\right\|\right] \\
&= \mathop{\mathbb{E}}_x\left[\left\|\widehat{\mathcal{A}}_i(x) \cdot \mathop{\mathbb{E}}_{b_j}\left[\overline{\mathcal{B}_k}(g_j(x,b_j)) - B\right]\right\|\right] \text{ (since } a_i, b_j \text{ don't overlap)} \\
&\le w^2\delta \cdot \binom{m-1}{i} \cdot \binom{m-1}{j} + 3\gamma^{i+j+2}. \text{ (by Theorem 29)}
\end{aligned}$$

Similarly for the case that $i \le \lceil k/2 \rceil, j > \lceil k/2 \rceil$ we can show that

$$\mathop{\mathbb{E}}_x\left[\left\|\mathop{\mathbb{E}}_y\left[(A_{x,y,i} - A)(B_{x,y,j} - B)\right]\right\|\right] \le w^2\delta \cdot \binom{m-1}{i} \cdot \binom{m-1}{j} + 3\gamma^{i+j+2}$$

by Theorem 28. Finally, note that the case $i, j > \lceil k/2 \rceil$ does not exist because $i + j \le k$.

Taking the summation of all the cases, we get

$$\begin{aligned}
&\mathop{\mathbb{E}}_x\left[\left\|\mathop{\mathbb{E}}_y\left[\mathcal{C}_k(x,y)\right] - AB\right\|\right] \\
&\le 2w^2\delta \cdot \left(\sum_{i+j=k}\binom{m-1}{i}\binom{m-1}{j} + \sum_{i+j=k-1}\binom{m-1}{i}\binom{m-1}{j} + \binom{m-1}{k}\right) \\
&+ (k+1)\cdot 9\gamma^{k+2} + k \cdot 9\gamma^{k+1} + 2 \cdot 3\gamma^{k+1} \\
&\le 4w^2\delta \cdot \binom{2m-1}{k} + (9k+9)\gamma^{k+2} + (9k+6)\gamma^{k+1} \\
&\le (10k+11)\gamma^{k+1} \\
&\le (11\gamma)^{k+1}.
\end{aligned}$$

Moreover, note that $AB = M_{0..2m}$, and the construction of $\mathcal{C}_k$ does not depend on the matrices $M_{[2m]}^{\{0,1\}}$. (See Section 1 for how the arithmetic operations in $\mathcal{C}_k(x,y)$ are translated back to operations on pseudo-distributions.) Therefore there exists a $(2m, w, (11\gamma)^{k+1})$-robust PRPD (G, ρ). □

Finally we analyze the seed length of the recursive construction, and present the main theorem.

Theorem 36. *There exists an explicit (n, w, ε)-robust PRPD (G, ρ) such that*

- $s_{\text{out}}(G, \rho) = O\left(\log(1/\varepsilon) + \log n \log(nw) \log\left(\frac{\log(1/\varepsilon)}{\log n}\right)\right)$
- $s_{\text{in}}(G, \rho) = O\left(\log(1/\varepsilon) + \log n \log(nw) \log\left(\frac{\log(1/\varepsilon)}{\log n}\right)\right)$
- $\mu(G, \rho) = \text{poly}(1/\varepsilon)$

Moreover, for every B the approximator $\mathcal{G}$ has the same corresponding pseudodistribution.

Proof. Let c be the constant such that for every $\varepsilon, \delta > 0$ there exists a (ε, δ)-sampler $g : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ such that $n = m + c\log(1/\varepsilon) + c\log(1/\delta)$ and $d = c\log(1/\varepsilon) + c\log\log(1/\delta)$, as guaranteed in Theorem 19. WLOG assume that n is a power of 2. Define $\gamma = 1/n^4$. For every $0 \le h \le \log n$, every $k \ge 0$, we will inductively prove that there exists a $(2^h, w, (11^h\gamma)^{k+1})$-robust PRPD $(G_{h,k}, \rho_{h,k})$ with the following parameters.

- If $k \le 1$, $s_{\text{out}}(G_{h,k}, \rho_{h,k}) \le h \cdot (3ck\log(n/\gamma) + 7c\log(w/\gamma))$
- If $k > 1$, $s_{\text{out}}(G_{h,k}, \rho_{h,k}) \le 4ck\log(n/\gamma) + (\lceil \log k \rceil + 1) \cdot h \cdot (10c\log(w/\gamma))$
- If $k \le 1$, $s_{\text{in}}(G_{h,k}, \rho_{h,k}) \le ck\log(n/\gamma) + 4c\log(w/\gamma)$
- If $k > 1$, $s_{\text{in}}(G_{h,k}, \rho_{h,k}) \le ck\log(n/\gamma) + h \cdot (4c\log(kw/\gamma))$
- $\mu(G_{h,k}, \rho_{h,k}) \le \max(1, \binom{2^h - 1}{k})$

We will write $s_{\text{out},h,k} = s_{\text{out}}(G_{h,k}, \rho_{h,k})$ and $s_{\text{in},h,k} = s_{\text{out}}(G_{h,k}, \rho_{h,k})$ for short. First consider the terminal case $2k \ge 2^h$ or $h = 0$. In this case we simply take $s_{\text{out},h,k} = 0$, $s_{\text{in},h,k} = 2^h \le 2k$ and $\mu(G_{h,k}, \rho_{h,k}) = 1$ s.t. $G_{h,k}(x, y, i) = y$ and $\rho_{h,k}(x, y, i) = 1$. For the other cases, we show that we can get the intended parameters by constructing $(G_{h,k}, \rho_{h,k})$ with the recursion in Theorem 35. Note that based on the induction hypothesis we can assume $\mathcal{G}_{a,h-1,k}$ and $\mathcal{G}_{a+2^{h-1},h-1,k}$ have exactly the same parameters, so we consider the parameter of $\mathcal{G}_{a,h-1,k}$ only. We have seen that the bound for $\mu(G_{h,k}, \rho_{h,k})$ is correct. First we show that the bound for $s_{\text{in},h,k}$ is correct. Recall that in the recursion we take parameters $d_i = c\log(1/\varepsilon_i) + c\log\log(1/\delta) \le ci\log(n/\gamma) + 2c\log(knw/\gamma)$, based on the fact that $\binom{2^h - 1}{i} \le n^i$. Now consider the restriction on $s_{\text{in}}(\mathcal{G}_k)$ in our recursion. For $i + j \le k$ and $j \le i \le \lceil k/2 \rceil$, we need

$$d_i + d_j \le ck\log(n/\gamma) + 4c\log(knw/\gamma) \le s_{\text{in},h,k}$$

which is true. For $i + j \le k$ and $i > \lceil k/2 \rceil$, we need

$$\begin{aligned} s_{\text{in},h-1,i} + d_j &\le ci\log(1/\gamma) + (h-1) \cdot 4c\log(inw/\gamma) + (cj\log(1/\gamma) + 2c\log(knw/\gamma)) \\ &\le ck\log(1/\gamma) + h \cdot 4c\log(knw/\gamma) \\ &\le s_{\text{in},h,k} \end{aligned}$$

which is also true. Moreover, observe that when $k \le 1$ it is always the case that $i, j \le \lceil k/2 \rceil$. Therefore the third condition is also true. Finally we show

that the bound for $s_{\text{out},h,k}$ is also correct. First observe that the restriction $s_{\text{out},h-1,i} \leq s_{\text{out},h,k}$ is trivially true. Then the only condition left is that for every $i \leq \lceil k/2 \rceil$,

$$s_{\text{out},h-1,i} + s_{\text{in},h-1,i} + c\log(1/\delta) + c\log(1/\varepsilon_i) \leq s_{\text{out},h,k}.$$

Since $s_{\text{out},h-1,i} \leq s_{\text{out},h-1,\lceil k/2 \rceil}$ and $s_{\text{in},h-1,i} \leq s_{\text{in},h-1,\lceil k/2 \rceil}$ for every i, it suffices to show that

$$s_{\text{out},h-1,\lceil k/2 \rceil} + s_{\text{in},h-1,\lceil k/2 \rceil} + c\log(1/\delta) + c\log(1/\varepsilon_{\lceil k/2 \rceil}) \leq s_{\text{out},h,k}.$$

First we consider $k \leq 1$, which is the case that $\lceil k/2 \rceil = k$. Then

$$\begin{aligned}
&s_{\text{out},h-1,\lceil k/2 \rceil} + s_{\text{in},h-1,\lceil k/2 \rceil} + c\log(1/\delta) + c\log(1/\varepsilon_{\lceil k/2 \rceil}) \\
&\leq s_{\text{out}}(\mathcal{G}_{a,h-1,k}) + 3ck\log(n/\gamma) + 7c\log(n/\gamma) \\
&\leq h \cdot (3ck\log(n/\gamma) + 7c\log(n/\gamma)) \\
&\leq s_{\text{out},h,k}.
\end{aligned}$$

Finally we consider the case $k > 1$. Observe that

$$\begin{aligned}
&s_{\text{out},h-1,\lceil k/2 \rceil} + s_{\text{in},h-1,\lceil k/2 \rceil} + c\log(1/\delta) + c\log(1/\varepsilon_{\lceil k/2 \rceil}) \\
&\leq s_{\text{out},h-1,\lceil k/2 \rceil} + s_{\text{in},h-1,\lceil k/2 \rceil} + \frac{3k+1}{2} \cdot c\log(n/\gamma) + 3c\log(w/\gamma) \\
&\leq s_{\text{out},h-1,\lceil k/2 \rceil} + (2k+1) \cdot c\log(n/\gamma) + (h-1) \cdot 4c\log(w/\gamma) + 7c\log(w/\gamma) \\
&\leq 4c \cdot \frac{k+1}{2} \cdot \log(n/\gamma) + \left(\lceil \log \lceil \frac{k}{2} \rceil \rceil + 1 \right) \cdot (h-1) \cdot (10c\log(nw/\gamma)) \\
&+ (2k+1) \cdot c\log(n/\gamma) + (h-1) \cdot 4c\log(w/\gamma) + 7c\log(w/\gamma) \\
&\leq 4ck\log(n/\gamma) + \left(\lceil \log \lceil \frac{k}{2} \rceil \rceil + 1 \right) \cdot (h-1) \cdot (10c\log(nw/\gamma)) + h \cdot 10c\log(nw/\gamma) \\
&\leq s_{\text{out},h,k}.
\end{aligned}$$

In the last inequality we use the fact that $\lceil \log k \rceil = \lceil \log(\lceil k/2 \rceil) \rceil + 1$ for every $k > 1$.

Finally, note that $(11^{\log n}\gamma) = n^{\log_2 11} \cdot n^{-4} \leq n^{-0.5}$. By taking $h = \log n$ and $k = \frac{\log(1/\varepsilon)}{\log(1/n^{0.5})}$, we get a (n, w, ε)-robust PRPD. □

Remark 37. *To get the seed length we claimed in Theorem 4, observe that the* $\log(1/\varepsilon)$ *term is dominating when* $\log(1/\varepsilon) \geq \log^3(nw)$*. Therefore we can simply replace the* $\log\log(1/\varepsilon)$ *factor on the* $O(\log n \log(nw))$ *term with* $\log\log(nw)$*.*

B Proof of Theorem 19

Lemma 38 (Theorem 19, restated. [18, 8])**.** *For every* $\delta, \varepsilon > 0$ *and integer* m*, there exists a* (ε, δ)*-sampler* $f : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ *s.t.* $d = O(\log\log(1/\delta) + \log(1/\varepsilon))$ *and* $n = m + O(\log(1/\delta)) + O(\log(1/\varepsilon))$*. Moreover, for every* x, y*,* $f(x, y)$ *can be computed in space* $O(m + \log(1/\delta) + \log(1/\varepsilon))$*.*

We will use the equivalence between seeded randomness extractor and oblivious sampler by Zuckerman [21]. To achieve the parameter we need, we need a

"high-entropy seeded extractor" such that the seed length only depends on entropy loss but not the length of source. We will use the standard "block-source" construction for high-entropy extractor which can be found in [10, 18]. For simplicity, we will use simple composition instead of zig-zag composition [18] because we are not aiming for optimal entropy loss. We will use the following standard lemmas for the extractor construction. Some of the following lemmas are implicit in their original source, and we recommend the readers to see [9, 20] for a proof.

Definition 39 ([6])**.** (X_1, X_2) *is a* (k_1, k_2)-block source *if* X_1 *is a* k_1*-source, and for every* $x_1 \in \mathrm{Supp}(X)$*,* X_2 *conditioned on* $X_1 = x_1$ *is a* k_2*-source.*

Lemma 40 ([10])**.** *Let* $X \in \{0,1\}^n$ *be a* $(n - \Delta)$ *source. Then for every integer* $0 \le t \le n$*,* X *is* ε*-close to a* $(t - \Delta, n - t - \Delta - \log(1/\varepsilon))$*-block source* (X_1, X_2) *where* $X_1 \in \{0,1\}^t$ *and* $X_2 \in \{0,1\}^{n-t}$*.*

Lemma 41 ([16])**.** *Let* $E_1 : \{0,1\}^{n_1} \times \{0,1\}^d \to \{0,1\}^{d_2}$ *be a* (k_1, ε_1) *extractor and* $E_2 : \{0,1\}^{n_2} \times \{0,1\}^{d_2} \to \{0,1\}^m$ *be a* (k_2, ε_2) *extractor. Define* $E((x_1, x_2), s) = E_1(x_2, E_2(x_1, s))$*. Then for every* (k_1, k_2)*-block source* $(X_1, X_2) \in \{0,1\}^{n_1} \times \{0,1\}^{n_2}$*,* $E((X_1, X_2), U_d)$ *is* $(\varepsilon_1 + \varepsilon_2)$*-close to uniform.*

Lemma 42 ([10])**.** *For every* $\varepsilon, \Delta > 0$ *and integer* n *there exists a* $(n - \Delta, \varepsilon)$ *extractor* $E : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^n$ *with* $d = O(\Delta + \log(1/\varepsilon))$*, and for every* x, y*,* $E(x, y)$ *can be computed in space* $O(n + \log(1/\varepsilon))$*.*

Lemma 43 ([9, 13])**.** *For every* $\varepsilon > 0$*, integer* $m > 0$ *and* $n \ge 2m$*, there exists a* $(2m, \varepsilon)$ *extractor* $E : \{0,1\}^n \times \{0,1\}^d \to \{0,1\}^m$ *with* $d = O(\log m + \log(1/\varepsilon))$*, and for every* x, y*,* $E(x, y)$ *can be computed in space* $O(m + \log(1/\varepsilon))$*.*

Lemma 44 ([21])**.** *Every* $(n - \log(1/\delta) - 1, \varepsilon)$*-extractor is a* (ε, δ)*-sampler.*

Now we show how to construct the sampler we need, and that it is indeed space efficient.

Proof. Let $\Delta = \log(1/\delta) + 1$. Let $E_1 : \{0,1\}^m \times \{0,1\}^{d_1} \to \{0,1\}^m$ be an $(m - \Delta, \varepsilon/3)$-extractor from Lemma 42, w.l.o.g. assume that $d_1 \ge \Delta + \log(3/\varepsilon)$. Then let $E_2 : \{0,1\}^{3d_1} \times \{0,1\}^d \to \{0,1\}^{d_1}$ be an $(2d_1, \varepsilon/3)$-extractor from Lemma 43. Then we claim that $E : \{0,1\}^{m+3d_1} \times \{0,1\}^d \to \{0,1\}^m$, defined as $E((x_1, x_2), s) = E_1(x_1, E_2(x_2, s))$, is a $(m + 3d_1 - \Delta, \varepsilon)$ extractor, and hence a (ε, δ) sampler by Lemma 44.

To prove the claim, consider any $(m + 3d_1 - \Delta)$-source X. By Lemma 40, X is $(\varepsilon/3)$-close to a $(m - \Delta, 3d_1 - \Delta - \log(3/\varepsilon))$-block source $(X_1, X_2) \in \{0,1\}^{3d_1} \times \{0,1\}^m$. By Lemma 41, $E_1(X_1, E_2(X_2, U_d))$ is $2\varepsilon/3$-close to uniform. Since $E(X, U_d)$ is $\varepsilon/3$-close to $E_1(X_1, E_2(X_2, U_d))$, by triangle inequality it is ε-close to uniform. Moreover, $d = O(\log(d_1/\varepsilon)) = O(\log\log(1/\delta) + \log(1/\varepsilon))$, $n = m + 3d_1 = m + O(\log(1/\delta) + \log(1/\varepsilon))$, and the required space to compute E is $O(m + d_1 + \log(1/\varepsilon)) = O(m + \log(1/\varepsilon) + \log(1/\delta))$. □

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